

A Star Schema-Based Cybersecurity Data Warehouse Framework for Knowledge Discovery from Phishing and Social Engineering Data Using OLAP and Data Mining Techniques

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Abstract—Phishing and social engineering threat data are typically scattered across disconnected sources with inconsistent structure, limiting systematic knowledge discovery. This paper presents a star-schema-based cybersecurity data warehouse for phishing and social engineering incident analysis, integrating three heterogeneous sources into a unified fact and dimension table. Online Analytical Processing (OLAP) operations, including roll-up, drill-down, slice, dice, and cube, were implemented directly in SQL and evaluated for query performance, achieving sub 2ms response times on filtered analytical queries and under 12ms on full multidimensional joins. A data mining layer was built on top of the warehouse: K-means clustering identified four structurally distinct phishing URL construction patterns, and fifteen classification algorithms were evaluated for phishing/benign URL prediction. CatBoost achieved the strongest overall performance of 95.35% accuracy, 96.91% precision, 98.07% recall, 97.49% F1-score, 96.99% ROC-AUC, outperforming other gradient-boosting, ensemble, and kernel-based classifiers, including Support Vector Machines. The proposed framework demonstrates that a lightweight, SQL-native OLAP and mining pipeline can support both descriptive and predictive cybersecurity analytics without specialized big-data infrastructure.

Keywords: cybersecurity, data warehouse, star schema, OLAP, phishing detection, social engineering, K-means clustering, CatBoost, ensemble classification

I. INTRODUCTION

Phishing and social engineering remain among the most persistent and evolving attack vectors in the cybersecurity landscape, exploiting human trust and behavioral vulnerabilities rather than purely technical weaknesses. Despite their prevalence, data describing these attacks is typically fragmented across disconnected sources threat-intelligence feeds, incident reports, and case-study archives each with inconsistent structure, variable metadata quality, and little interoperability. This fragmentation limits the ability of researchers and practitioners to perform systematic, longitudinal analysis of phishing and social engineering trends, and constrains the application of analytical techniques such as OLAP and data mining that require a unified, queryable data model.

Raw threat intelligence feeds compound this challenge: many widely used sources (e.g., community phishing-URL

feeds) provide minimal contextual metadata, often limited to a bare URL with no associated timestamp, target information, or severity indicator. This metadata poverty restricts the depth of analysis possible on otherwise large-volume data sources, and necessitates feature derivation and integration strategies when such sources are combined with richer, but smaller, structured datasets.

This paper addresses these limitations by designing and implementing a star schema-based data warehouse for phishing and social engineering knowledge discovery. The proposed system integrates heterogeneous data sources a labeled phishing/benign URL dataset with real world timestamps and target-brand metadata, a high volume phishing URL feed, and a hand-curated social engineering case archive into a unified fact-and-dimension model, enabling structured OLAP based trend analysis and downstream data mining. Guided by the systematic literature review, this study is structured around three research questions: (1) How can a star schema-based data warehouse be designed to efficiently store and organize phishing and social engineering data for cybersecurity analysis? (2) What OLAP techniques can be applied to the data warehouse to identify meaningful patterns and trends in phishing and social engineering attacks? (3) Which data mining techniques integrated with the star schema warehouse can best predict and classify phishing and social engineering threats for knowledge discovery?

The key contributions of this research include the development of a star-schema based data warehouse with a reproducible ETL pipeline based on the integration of three heterogeneous data sources. A native SQL OLAP layer performs the roll-up, drill-down, slice, dice and cube operations, including performance testing. The data mining layer includes the clustering of phishing URL features using K-means clustering and the classification using fifteen different classifiers, where CatBoost classifier has performed best.

II. RELATED WORK

Phishing detection has been extensively studied using both classical machine learning and deep learning approaches. Feature based classifiers using URL and HTML charac-

teristics remain widely used for their interpretability and low computational cost [1]–[4], while more recent work has shifted toward deep learning and transformer-based architectures, including CNN attention hybrids [5], CNN LSTM frameworks [6], and large language model-based detectors [7], [8]. Systematic reviews of this space [9]–[13] consistently report a trade-off between the predictive strength of deep models and the interpretability of lightweight, feature-based classifiers a tension this study addresses directly by evaluating both a structural feature SVM and an ensemble of fifteen classifiers within a single, warehouse integrated pipeline.

Large-scale, measurement-driven phishing studies most closely inform the data acquisition strategy of this work. Choo et al. [14] analyzed VirusTotal reports at scale to characterize phishing and malware URLs, while Moura et al. [15] studied phishing at the ccTLD level to quantify mitigation effectiveness. Prasad and Chandra [16] introduced a diverse, similarity index based phishing URL dataset (PhiUSIIL) that, like the datasets integrated in this paper, prioritizes real-world metadata richness over purely synthetic feature engineering.

Social engineering has received comparatively less data-driven treatment. Existing work is largely survey or taxonomy based, characterizing attack psychology [17], defense mechanisms [18], [19], and prevention strategies [20], [21], with generative AI driven social engineering emerging as a recent concern [22]. Notably, none of this literature integrates social engineering case data into a structured, queryable warehouse alongside phishing-URL data a gap this paper addresses.

Closest in spirit to the present work is research on knowledge discovery from cyberthreat intelligence. Rahman et al. [23] mined temporal attack patterns from threat intelligence reports, and Molina-Coronado et al. [24] surveyed network intrusion detection through the lens of the knowledge discovery in databases (KDD) process. However, neither applies a dimensional data warehouse model with native OLAP support to phishing/social engineering data specifically the contribution this paper makes.

III. PROPOSED METHOD

The proposed framework follows a sequential architecture of analytical processing, whereby each step prepares or enhances the data for the following one. In the first step, the data warehouse is developed as the primary analytical repository. Second, the OLAP techniques are applied to analyze the stored data from various dimensions. The third step implements unsupervised clustering to reveal patterns in phishing URLs. Finally, supervised classification is used to classify URLs as either benign or phishing. This architectural design segregates the tasks of data structuring, multidimensional analysis, pattern recognition, and predictive modeling in an integrated manner within an analytical process. Figure 1 describes the proposed framework employed in this study.

A. Star Schema-Based Data Warehouse

A Star Schema was used as the dimensional model of storing the data about cybersecurity incidents. The star schema includes the central **FACT_INCIDENT** table, embodying main analytical attributes of each incident,

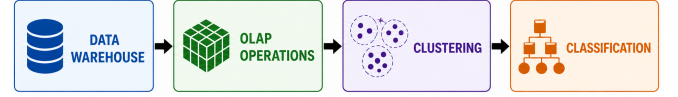


Fig. 1: Proposed Cybersecurity Data Mining Framework

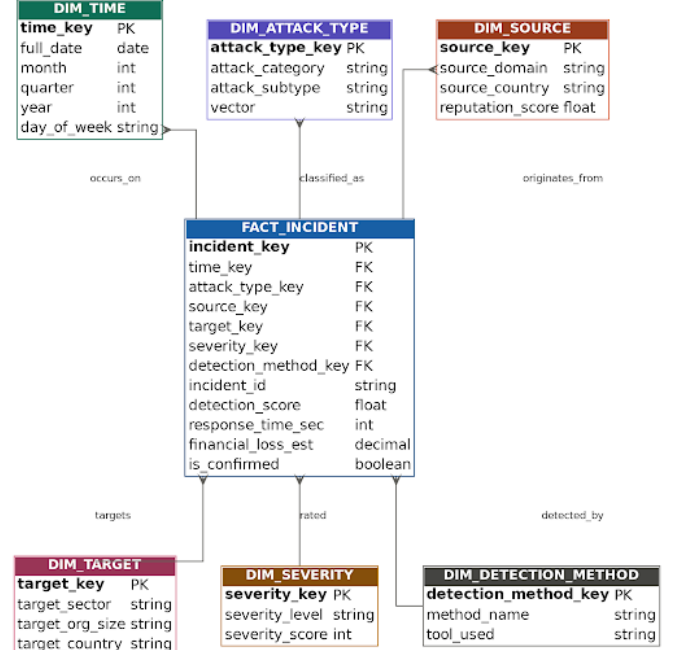


Fig. 2: Star Schema-Based Data Warehouse for Cybersecurity Incidents

which is surrounded by six dimension tables: **DIM_TIME**, **DIM_ATTACK_TYPE**, **DIM_SOURCE**, **DIM_TARGET**, **DIM_SEVERITY**, and **DIM_DETECTION_METHOD**, as shown in Figure 2. The fact table is the central entity of the warehouse and comprises of foreign key links for each incident in relation to the dimensions, together with other measures and attributes related to an incident such as detection score, response time, financial loss, and confirmation status. Thus, using the dimensional approach, an incident can be analyzed simultaneously along all temporal, categorical, geographical, organizational, severity, and detection attributes. The star schema therefore forms the basis for OLAP operations.

B. OLAP-BASED CYBERSECURITY ANALYSIS

After the data warehouse creation, six OLAP operations such as aggregation, slice, dice, roll-up, drill-down, and cube were conducted for the analysis of cyber security incidents from multiple dimensions.

Aggregation by severity: The severity aggregation resulted in 6,307 unknown severity incidents, 274 incidents with medium severity, 135 with high severity, and 10 critical incidents. The prevalence of unknown severity points out the incompleteness of severity data and limits its detailed-based analysis.

Slice Operation: A slice was performed on the phreshphish data set and categorized according to attack subcategory. There were 264 generic phishing and 142 brand impersonation attacks. This is an example of how source dimension-based techniques can be used for examining particular sources of threat.

Dice Operation: The dice analysis involved combining various criteria based on source, date, type of attack, sub-type of attack, and severity level. The analysis revealed that there were 264 medium level of severity generic phishing attacks, 132 high level of severity brand impersonation attacks, 10 critical level brand impersonation attacks, and several social engineering attacks. Multidimensional filtering gives a more comprehensive insight into relationships between attack attributes.

Roll-Up Analysis: Roll-up analysis was conducted hierarchically in terms of time starting from months and then moving on to years. The data set based on the particular dates included 273 cases in 2024, 142 cases in 2025, and four cases in 2026, which makes the total number of 419 cases. This explains how temporal hierarchies enable the use of analysis at various levels of abstraction.

Drill-Down Analysis: A drill-down analysis was conducted in July 2024 to conduct a deeper investigation into the results that had been rolled up before. The analysis found out 111 phishing events, which shows how roll-up and drill-down can be used complementarily to navigate through temporal patterns.

Cube Analysis: Cube Analysis included the combination of attack severity and attack category resulting in 406 cases of phishing attacks and 13 cases of social engineering attacks, totaling 419 cases in the specific date dataset. In summary, the OLAP analysis shows how the proposed data warehouse allows conducting flexible analysis regarding time, severity, source, attack category, and type of attack.

C. DATA MINING THROUGH CLUSTERING

After performing the OLAP mining operations, K-means clustering was employed as an unsupervised data mining method to detect the underlying structure of the phishing URLs without depending on any predefined class labels. The number of clusters was determined for $k = 2$ to $k = 10$ through the elbow method and silhouette coefficient techniques. In the elbow method, the within-cluster variation was analyzed while in the silhouette coefficient technique, cluster cohesion and separation were analyzed. The proposed solutions were later validated with respect to interpretability and the structure of the detected clusters, and the chosen clustering solution was then used to define different phishing URL patterns.

D. SUPERVISED CLASSIFICATION OF Phishing URLs

In addition to unsupervised pattern detection, supervised Machine Learning classification was also used to classify between benign and phishing URLs. The classification process is the predictive part of the proposed framework. This process includes several successive steps such as data preprocessing, feature selection, classifier training, model evaluation, model selection/tuning, and labeling. Figure 3 illustrates the proposed framework for the URL classification.

1) Data Preprocessing: Data preprocessing is a crucial step in ensuring the quality of the data and its compatibility with Machine Learning algorithms is improved. The first step in the classification process is preparation of the mining data for the purpose of analysis by machine learning algorithms. preprocessing includes handling missing values, discarding

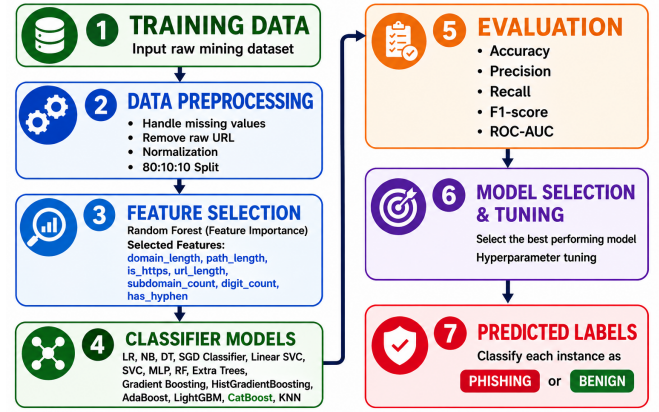


Fig. 3: Proposed Machine Learning Classification Framework for Phishing Detection

the raw URL attribute, normalization of data, and partitioning of the data set in the 80:10:10 ratio. The data set contains no missing values and all are in numeric form. The subsets of training, validation, and testing are used for training of the model, evaluation of performance of the algorithm during development of the model, and evaluation of performance of the trained model, respectively.

2) Feature Selection: After the preprocessing step, the Random Forest feature importance ranking was performed for feature selection. This step aims at selecting those features that add value to the classification task, without including any redundant input features. The top seven URL features performed very well and were therefore chosen for the next step, which involved classification. The selected features of the URLs based on the classification process include the following: domain length, path length, HTTPS, URL length, subdomain number, digit number, and the existence of hyphens. These features are used to construct a numeric representation of the URL's structural features and can then be processed using a classifier.

3) Classification Models: Several classification techniques have been implemented to understand the predictive capabilities of different machine learning techniques. It included classifiers, such as Logistic Regression (LR), K-Nearest Neighbors (KNN), Decision Tree (DT), Stochastic Gradient Descent (SGD) Classifier, Support Vector Classifier (SVC), Linear Support Vector Classifier (Linear SVC), Multi-Layer Perceptron (MLP), Random Forest (RF), Extremely Randomized Trees (Extra Trees), Gradient Boosting Classifier, Histogram-based Gradient Boosting (HistGradientBoosting), Adaptive Boosting (AdaBoost), Categorical Boosting (CatBoost), and Light Gradient Boosting Machine (LightGBM). From these candidate models, five algorithms, which have performed well were selected for comparative analysis and those are CatBoost, Random Forest, XGBoost, LightGBM, and MLP. The models employ different learning techniques such as tree-based algorithms, gradient boosting techniques, and neural networks to classify data. After performing comparative analysis, CatBoost is selected as the model and it is explained in the Results section.

In the CatBoost model, an ensemble of decision trees is formed iteratively, where the next decision tree helps in correcting errors committed by the previous ensemble.

CatBoost uses ordered boosting technique to minimize prediction shift and overfitting, and the symmetric tree structure of CatBoost uses the same split criteria at every depth of a tree. For the current task of classification, CatBoost was trained using selected numerical URL features in order to capture nonlinear dependencies between them, which are related to phishing and benign URLs. Tuning of hyperparameters was done to find optimal configuration of the CatBoost classifier. Hyperparameters include the following: number of iterations (iterations), depth of a tree (depth), learning rate (learning_rate), L2 regularization (l2_leaf_reg), and feature binning (border_count). Finally, 200 boosting iterations, tree depth 10, learning rate of 0.05, L2 regularization coefficient of 3, and border count of 32 were chosen as hyperparameters in the CatBoost classifier configuration.

4) *Performance Evaluation*: The accuracy, precision, recall, F1-score and ROC-AUC were used for evaluation of the classifiers. Accuracy reflects the percentage of accurately classified samples. Precision refers to the percentage of positive classifications that are accurate, while recall reflects the capability of the classifier to detect the positive class samples. F1-score reflects the harmonic mean of precision and recall and hence balances between these two. ROC-AUC allows another way to evaluate the performance of the classifiers in terms of distinguishing between the classes at different classification levels. The use of the aforementioned metrics allows the comparison of the models from different angles.

IV. EXPERIMENTS AND EVALUATION

A. Dataset Collection

The classification data set contains a total of 7,307 URL records, out of which 6,713 are phishing URLs, and 594 are benign URLs. Phishing URLs were collected by combining 406 records from phreshphish and 6,307 records from the OpenPhish Kaggle dataset, while the benign URLs were collected from phreshphish. Berkeley Social Engineering Dataset was excluded from the classification process because it lacked the URL structure needed to extract the features used in the classification process. Seven numerical structural features are used to represent each URL for the classification process: URL length, path length, number of subdomains, HTTPS existence, domain length, hyphen existence, and digit count. Domain Length and Path Length show the length of individual URL elements, whereas URL Length shows the overall structural characteristic. HTTPS shows whether the secure protocol is used. Subdomain Count shows the complexity of the domain hierarchy, Digit Count the presence of numeric characters, and Hyphen Indicator shows whether there are hyphens in the URL.

B. Results and Analysis

1) *Feature Selection Analysis*: The feature importance analysis reveals that the domain length has been determined as the most important factor in the classification of phishing URLs, with a Mean Decrease in Gini of around 0.206, while the second highest is path length (0.173), and the third is HTTPS (0.154). The length of the URL (0.149) and the number of subdomains (0.146) play a significant role, and the number of digits (0.088) and presence of a hyphen (0.084) are the least significant. It can be inferred

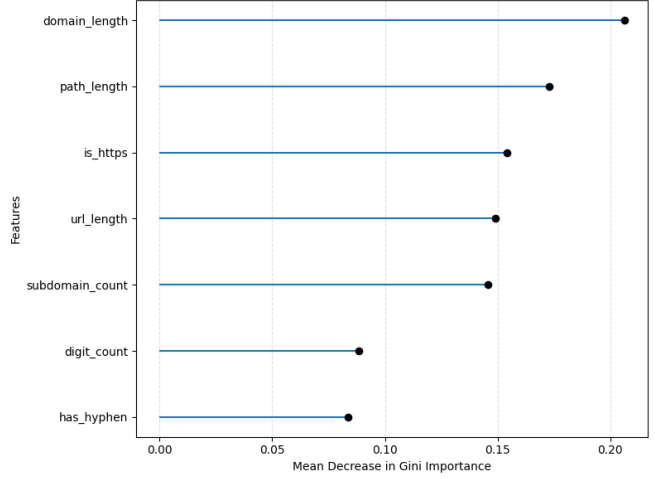


Fig. 4: Random Forest Feature Importance based on Mean Decrease in Gini

TABLE I: Comparative performance of the evaluated models

Model	Accuracy	Precision	Recall	F1-Score
CatBoost	0.9535	0.9691	0.9806	0.9748
Random Forest	0.9522	0.9690	0.9806	0.9741
XGBoost	0.9507	0.9676	0.9792	0.9734
LightGBM	0.9507	0.9676	0.9792	0.9734
MLP	0.9494	0.9662	0.9792	0.9726

from the findings that the URL structure contributes more in classifying phishing URLs. Figure 4 illustrates the feature importance ranking using Random Forest.

2) *Performance Without SMOTE*: Without applying Synthetic Minority Over-sampling Technique (SMOTE), CatBoost achieved the best overall performance, with an accuracy of 95.35%, precision of 96.91%, recall of 98.06%, and F1-score of 97.48%. RF also performed competitively, achieving an F1-score of 97.41%. Overall, the classifiers achieved F1-scores above 97%, demonstrating strong phishing-URL classification performance. Table I summarizes the performance of 5 top performing models.

The confusion matrix as shown in Figure 5, illustrates that the model accurately predicted 38 benign and 659 phishing URLs, but 21 benign URLs were misclassified as phishing and 13 phishing URLs were incorrectly predicted as benign. This means that the total number of correct predictions by the model is 697 out of 731 samples, which is equivalent to 95.35% accuracy. Since the false negative rate is quite low (13), this implies that the model successfully identifies most of the phishing URLs but has some misclassifications of benign URLs.

The ROC curves show high discrimination performance for all five models with AUC measures from 0.9670 up to 0.9756 as shown in Figure 6. The curves are located much higher than the random classifier threshold (AUC = 0.5000), which means that the models can discriminate phishing from benign URLs efficiently. Despite the fact that MLP obtained the highest ROC-AUC (0.9756), CatBoost was chosen due to its good classification results and balanced measures for different evaluation methods. This model managed to reach 95.35% accuracy, 96.91% precision, 98.06% recall and 97.48% F1-Score. That shows that this classifier can detect

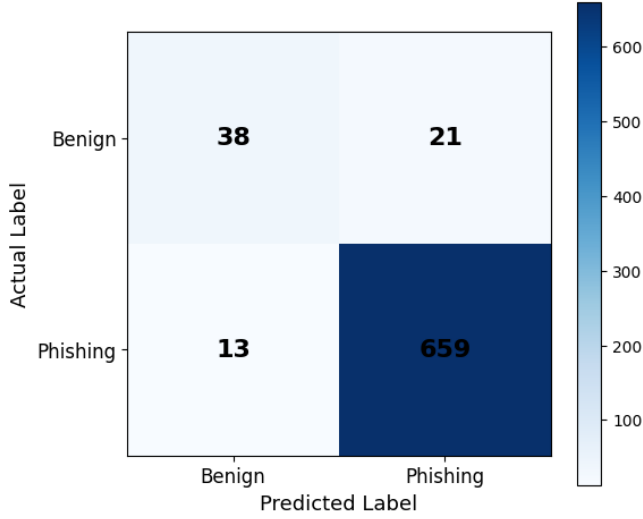


Fig. 5: Confusion Matrix

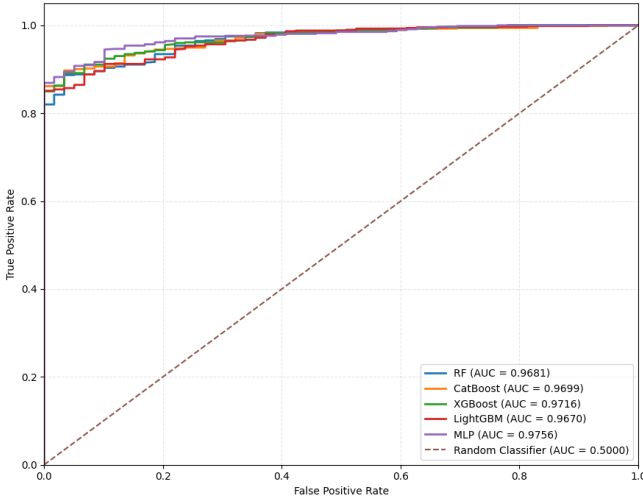


Fig. 6: ROC Curves of Top Classifiers Without SMOTE

the phishing URLs and does not have too many false-positive results.

3) *Effect of SMOTE*: The impact of SMOTE was studied using CatBoost as a means to balance out any possible class imbalance present in the phishing URL data. By applying SMOTE, synthetic examples are created for the minority class through interpolation of minority class samples. Without SMOTE, CatBoost achieved an F1-score of 97.48%, whereas applying SMOTE resulted in an F1-score of 95.41%. Although precision increased from 96.91% to 98.11%, accuracy and recall decreased from 95.35% to 91.79% and from 98.06% to 92.86%, respectively. Thus, SMOTE improved precision but reduced the overall performance within the experimental setting. As a result, the use of SMOTE algorithm has not helped to enhance the overall performance of the classifier in this case, resulting in the original class distribution being retained as the final classification model.

C. Clustering Results and Analysis

The Elbow and Silhouette methods were used to determine the appropriate number of clusters. Although the Silhouette Method showed a higher score for some larger values of k , $k = 4$ was selected because it produced more interpretable phishing-URL patterns. In contrast, $k = 9$ resulted in poorly

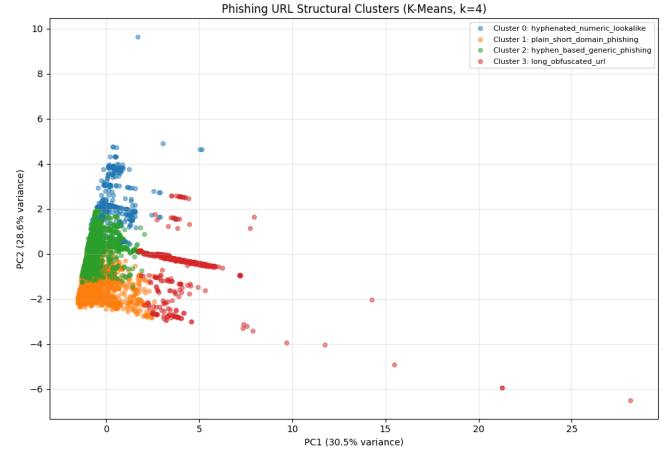


Fig. 7: Visualization of the Four Identified Phishing-URL Clusters

interpretable clusters and a small outlier cluster. The $k = 4$ clustering solution identified four distinct phishing-URL patterns: hyphenated numeric lookalike URLs ($n = 749$), plain short-domain phishing URLs ($n = 2163$), hyphen-based generic phishing URLs ($n = 3263$), and long obfuscated URLs ($n = 538$) as illustrated in Figure 7. These clusters demonstrate the diversity of structural patterns observed in phishing URLs.

V. COMPARATIVE ANALYSIS OF RESULTS

The comparison in Table II reveals that the proposed methodology achieves an accuracy level of 95.35% using only seven features and 7,307 samples, but there are some existing solutions with higher accuracy rates, which are provided by use of either more complicated feature representation or larger number of samples. Prasad and Chandra [16] got 99.79% with 54 features and 235,795 samples, whereas Xiao et al. [5] got 99.84% with character level CNN and multi-head self-attention with 88,984 samples. Similarly, accuracies higher than 98% have been obtained by Das Gupta et al. [2] and Opara et al. [3], with hybrid features, HTML information, and deep learning architecture respectively. Unlike all the previous approaches, the proposed framework uses only seven URL based features using CatBoost, along with data warehousing and OLAP based analytics. Therefore, the significance of the framework is not limited to just classification accuracy, but also in terms of being lightweight and integrated for knowledge discovery.

VI. CONCLUSION AND FUTURE WORK

This study developed an integrated cybersecurity data warehouse and data mining framework that combines a star-schema warehouse, OLAP analysis, clustering, and machine learning for cybersecurity and phishing-URL analysis. The warehouse enabled structured data management, while OLAP operations revealed patterns across attacks, severity, sources, and time. Clustering identified four distinct phishing-URL patterns, and CatBoost achieved the best classification performance with an F1-score of 0.9748. Future work will focus on integrating larger and more diverse datasets, developing richer phishing-URL features, and evaluating advanced classification and pattern-discovery models

TABLE II: Comparison of the Proposed Approach with Existing Phishing Detection Methods

Method / Work	Problem Dealt With	Features	Samples	Classification Model	Accuracy
Proposed Framework	Star schema-based cybersecurity data warehouse for phishing detection	7	7,307	CatBoost	95.35%
Prasad & Chandra [16] (PhiUSIIL)	Phishing URL detection via similarity index and incremental learning	54	235,795	Incremental / pre-trained classifier	99.79%
Das Gupta et al. [2]	Hybrid URL and hyperlink feature-based phishing website detection	25	NM ¹	XGBoost	99.17%
Xiao et al. [5] (CNN-MHSA)	Phishing website detection from URL strings	Char-level	88,984	CNN + multi-head self-attention	99.84%
Opara et al. [3] (WebPhish)	Phishing web page detection using raw URL and HTML	Raw embedding	NM ¹	End-to-end CNN	98.10%

¹NM = Not mentioned in the source paper/abstract.

to improve the scalability, generalizability, and effectiveness of the proposed framework.

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